

Figure 1: The main components of Samba

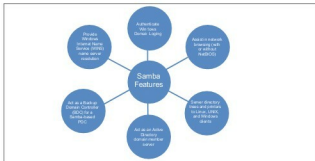


Figure 2: Features of Samba

Why use Samba?

Here are a few reasons:

- Samba easily replaces a Windows NT Samba file/print server.
- It offers more stability/performance compared to Windows NT.
- It allows interoperability between UNIX and Windows workstations.
- It can serve as a Windows primary domain controller.
- Samba can serve as a highly stable WINS server.
- It can be used to stabilise Windows browsing services.

Installation and configuration

Samba installation

To install Samba on a Linux machine, use the following command:

```
yum -y install samba
```

To create a new Samba user (the new Samba user must exist on the Linux machine as a user), use the following command:

```
smbpasswd -a <username>
```

For example, to add the user *max*, execute the command given below:

```
smbpasswd -a max
```

To create a Samba users (*smbgrp*) group, execute the following command:

```
groupadd smbgrp
```

To add a user into a group, use the command given below:

```
usermod -G smbusers max
```

Samba configuration

The *smb.conf* file stores all the configurations for Samba. It provides runtime configuration information for the Samba programs. It determines which system resources you can share with the outside world and what restrictions you wish to place on them. The file consists of sections and parameters. The name of the section must be enclosed in square brackets. Each section continues until the next one begins.

Each section of the file starts with a section header such as `[global]`, `[homes]`, `[printers]`, etc.

The `[global]` section has the variables to define sharing for all resources.

The `[homes]` section allows remote users to access their (and only their) home directory on the local (Linux) machine. That is, users can access their share from Windows machines and they will have access to their personal home directories.

The `[printers]` section works the same as the `[homes]` section, but for printers.

If you change this configuration file, the changes do not apply until you restart the Samba daemon with the command *smb restart*.

To view the status of the Samba daemon, use the following command:

```
/sbin/service smb status
```

To start the daemon, use the following command:

```
/sbin/service smb start
```

To stop the daemon, use the following command:

```
/sbin/service smb stop
```

To start the smb service at boot time, use the following command:

```
/sbin/chkconfig --level 345 smb on
```

Access management using Samba

Table 1 gives the access control options.